

TellCo: User Analytics in the Telecommunication Industry

Problem statement, solution and step-by-step method

Nexthikes IT Solutions - Data Science project
Data: one month of xDR sessions (150,000 rows, 55 columns)
Report date: 1 October 2026

Deliverable	File
Python code with all charts inside	TellCo_Analysis.ipynb
Presentation (17 slides)	TellCo_Presentation.pptx
This report (problem, solution, method)	TellCo_Project_Report.pdf

1. The problem

An investor wants to buy **TellCo**, a mobile service provider in the Republic of Pefkakia. TellCo's owners shared their financial numbers, but nobody has ever analysed the data that the network generates. The investor needs to know **whether TellCo is worth buying** and **which products or customer groups can increase profit**.

Our job as the data scientist: analyse a month of xDR (data session) records and answer four questions:

- **User overview** - which handsets and manufacturers do customers use, and how do they use apps?
- **User engagement** - how often, how long and how much do customers use the network?
- **User experience** - how good is the network quality (delay, retransmission, speed) for each customer and handset?
- **User satisfaction** - combining engagement and experience, who is satisfied, and can we predict it?

The result must be a clear recommendation (buy or not), supported by data, plus the limitations of the analysis.

2. The data

The file has 150,000 xDR sessions and 55 columns for 106,856 unique customers (identified by MSISDN). Columns cover: session start/end/duration, IMSI/MSISDN/IMEI, location, round-trip time (RTT), TCP retransmission, throughput, handset manufacturer/type, and download/upload bytes for 7 applications (Social Media, Google, Email, YouTube, Netflix, Gaming, Other).

Data problems we met

Problem found	How we solved it
One completely blank row at the end of the file	Removed it before any calculation.
TCP retransmission 59-64% missing, RTT about 19% missing, many "Nb of sec" columns 60-87% missing	Numeric gaps filled with the column mean, text gaps (handset) with the mode, as required by the brief. Columns with the highest gaps were not used as features.
Extreme outliers (e.g. RTT up to 96,923 ms, TCP up to 4.3 billion bytes)	Values outside $Q1 - 3 \times IQR$ and $Q3 + 3 \times IQR$ were replaced by the mean of the normal values. We used $3 \times IQR$ (not 1.5) so only truly extreme values are touched.
6.4% of sessions have handset "undefined" / missing	Treated as unknown and excluded from handset rankings, so "undefined" is not shown as a top manufacturer.
Application bytes add up to about 1.85 times Total DL+UL in every session	Cannot be fixed. We kept the numbers as given, used Total DL+UL for engagement and clearly flagged this as a limitation.
A few MSISDN values are not valid phone numbers (about $8.8e14$)	Kept (very few), noted as a data-entry issue.

3. Solution - step by step

Task 1 - User overview

What we did: counted sessions per handset and manufacturer; aggregated per customer the number of sessions, total duration, DL, UL and bytes per application; then ran a full exploratory analysis (data types, mean/median, dispersion, histograms, boxplots, app-vs-total scatter plots, duration deciles, correlation matrix and PCA).

- **Top handsets:** Huawei B528S-23A (19,752 sessions; a home 4G router), iPhone 6S (9,419), iPhone 6 (9,023), iPhone 7 (6,326), iPhone SE (5,187).
- **Top 3 manufacturers:** Apple 59,565 (39.7%), Samsung 40,839 (27.2%), Huawei 34,423 (23.0%).
- **Marketing advice:** use upgrade offers for old iPhones, create a home-internet package for router users and handset-linked data bundles for Samsung.

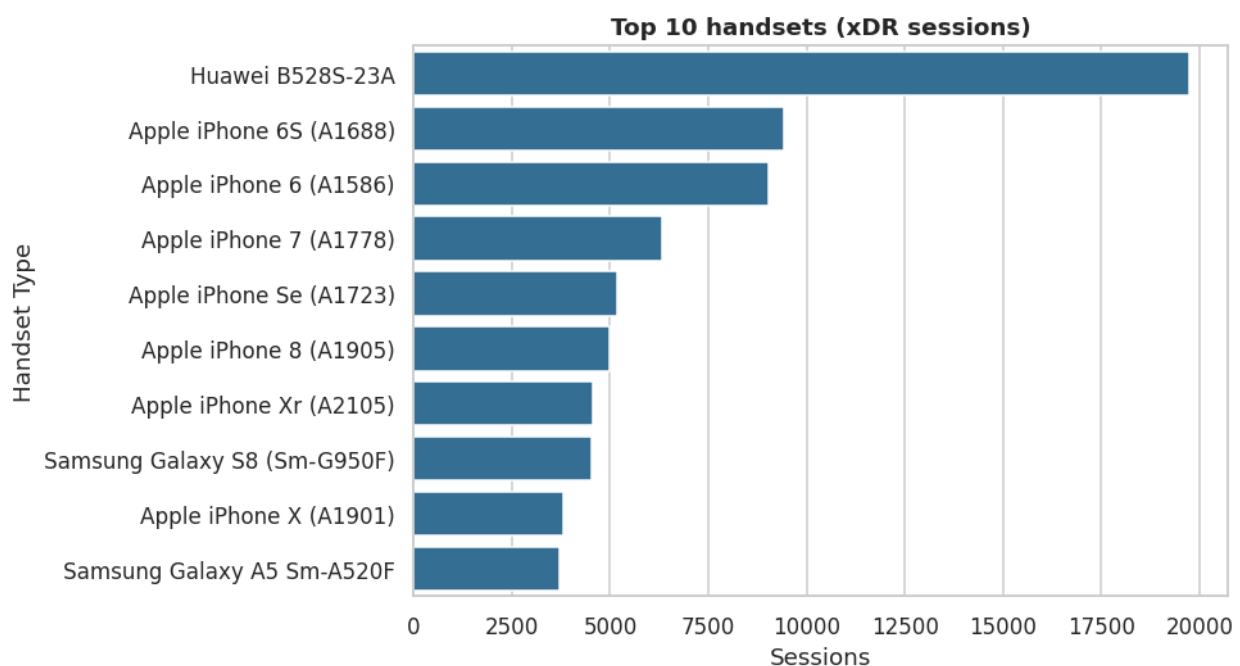


Figure 1 - Top 10 handsets by number of xDR sessions

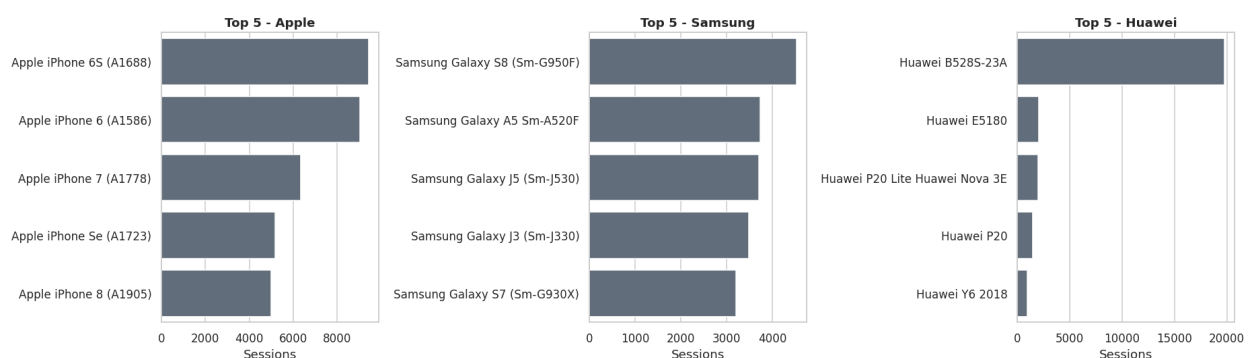


Figure 2 - Top 5 handsets for each of the top 3 manufacturers

Dispersion and distribution: almost all usage variables have a high coefficient of variation and positive skew: a small group of heavy users pulls the mean above the median. The median therefore describes the typical customer better. The full tables are printed inside the notebook.

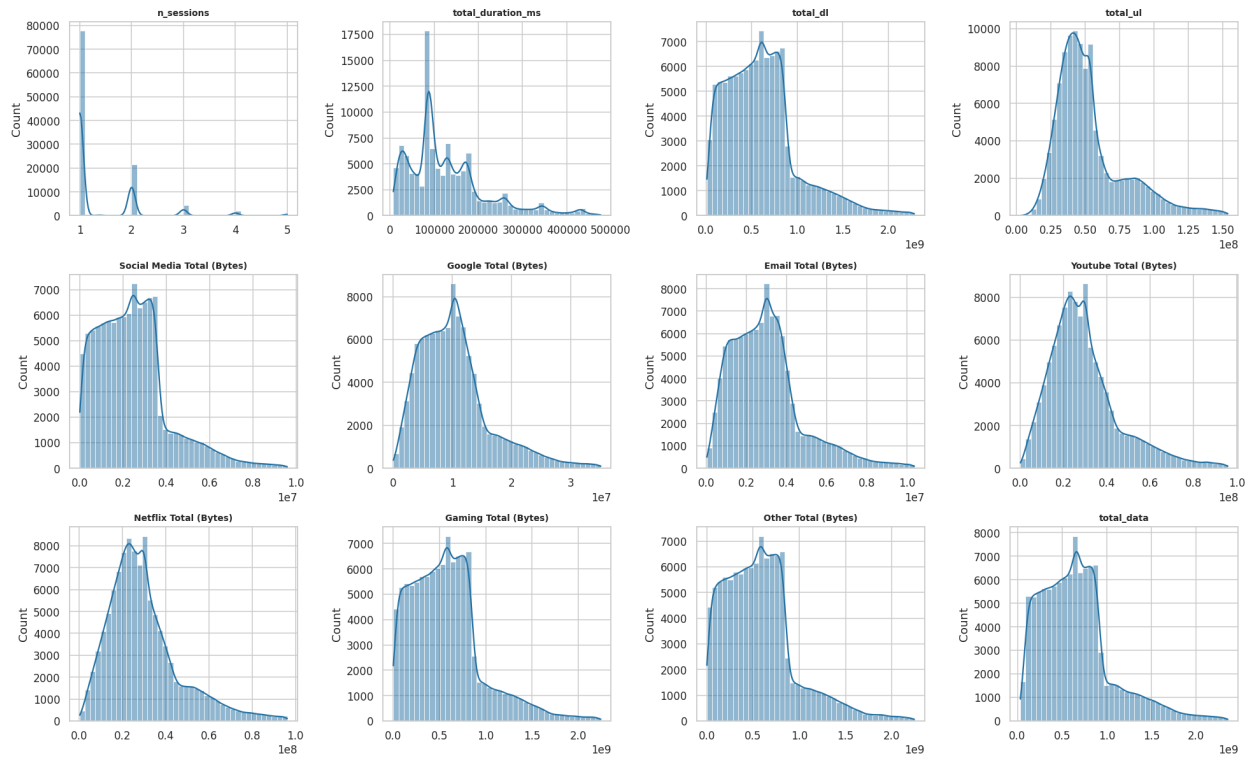


Figure 3 - Histograms of all user-level variables (after cleaning)

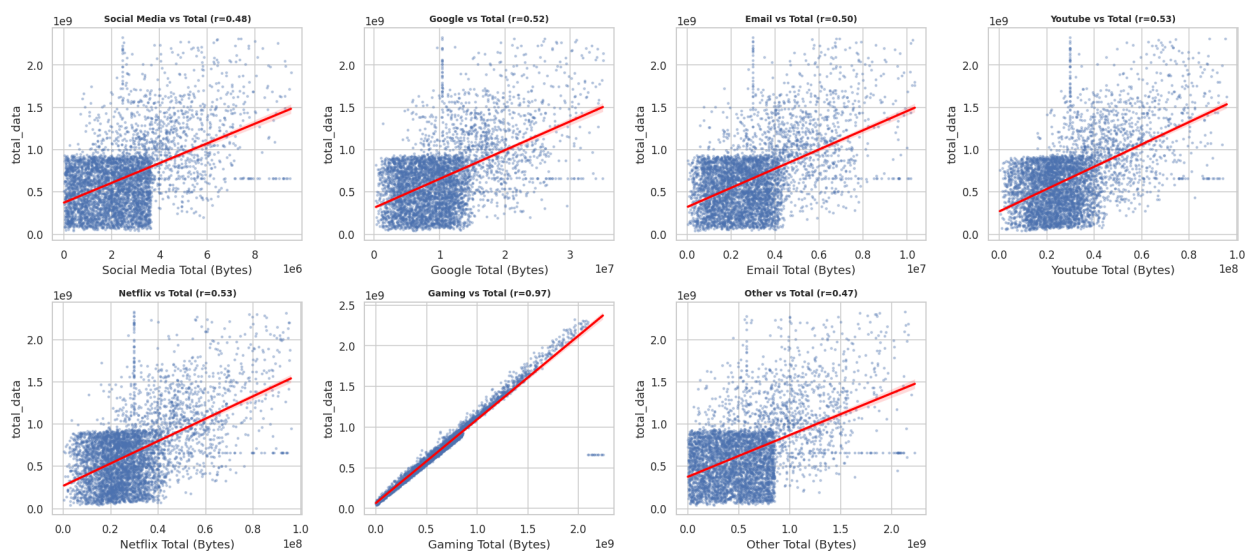


Figure 4 - Each application vs total DL+UL data (bivariate analysis)

Deciles: users were split into 10 groups by total session duration. The top five deciles (D6-D10) carry 58.6% of all data; the longest-duration decile (D10) alone carries about 10.4 TB.

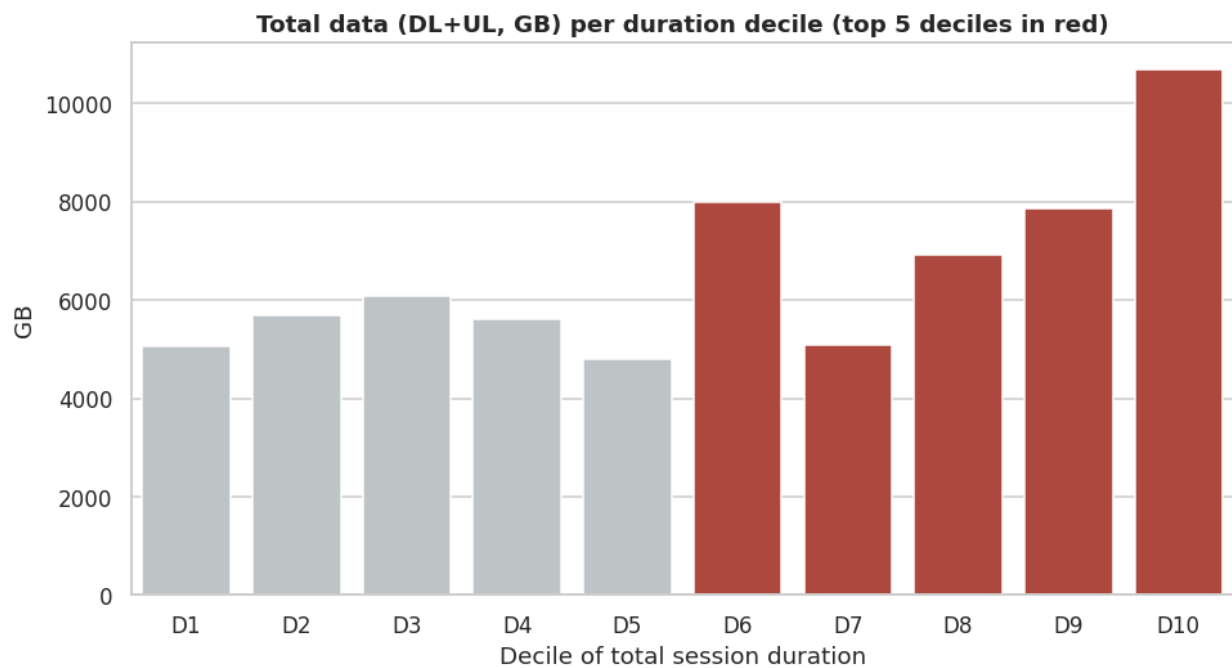


Figure 5 - Total data per duration decile

Correlation: all 7 application volumes are positively correlated at a moderate level (r about 0.45-0.50). Heavy users are heavy on every app, and no two apps are redundant.

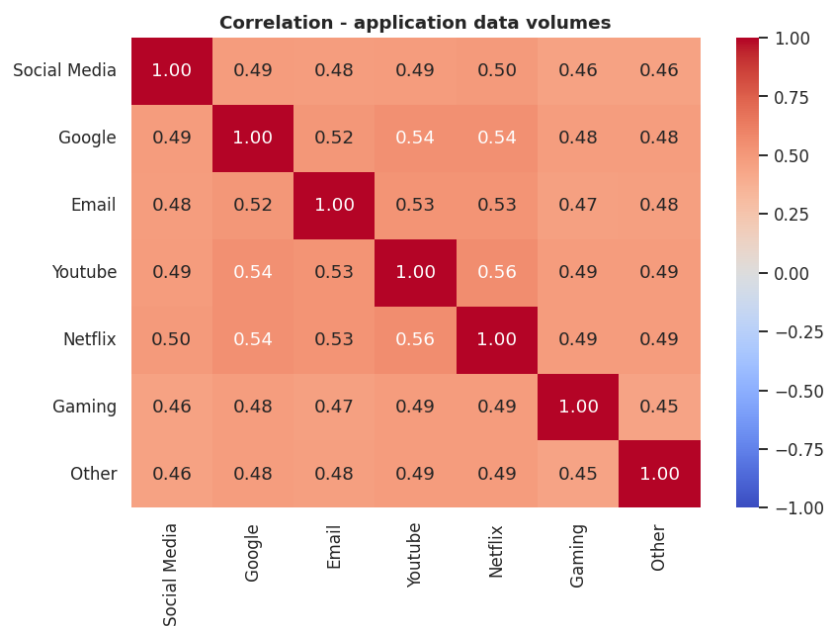


Figure 6 - Correlation between application data volumes

PCA (4 points): (1) PC1 explains 56.7% of the variance with almost equal loadings on all apps - a general heavy-user factor. (2) PC2 (7.9%) is driven by Gaming (+0.91), PC3 (7.7%) by Other (+0.87). (3) Three components explain about 72% and five about 87%, so the 7 dimensions can be reduced moderately. (4) PC1 is useful for segmentation and PC2 for targeting gamers.

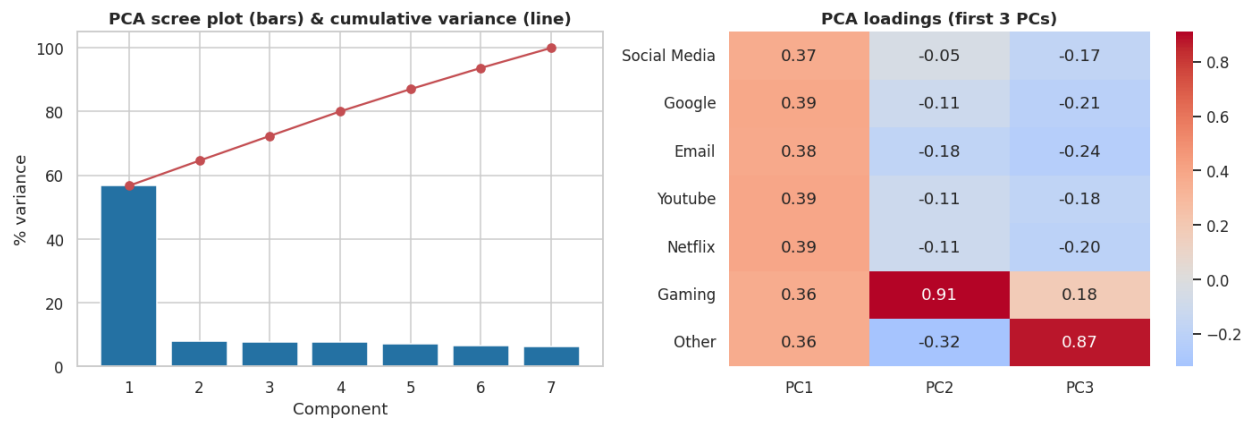


Figure 7 - PCA scree plot and loadings

Task 2 - User engagement

What we did: aggregated three engagement metrics per MSISDN (number of sessions, total duration, total traffic DL+UL); listed the top 10 customers for each metric; min-max normalised the metrics; ran k-means with k=3; computed min, max, mean and total of the original (non-normalised) values per cluster; found the top 10 users for each application; used the elbow method to choose k.

Cluster	Users	Avg sessions	Avg traffic	Share of traffic
Low engagement	81,163 (76%)	1.0	0.5 billion bytes	54.5%
Medium engagement	21,645 (20%)	2.2	1.1 billion bytes	33.0%
High engagement	4,048 (3.8%)	4.3	2.3 billion bytes	12.5%

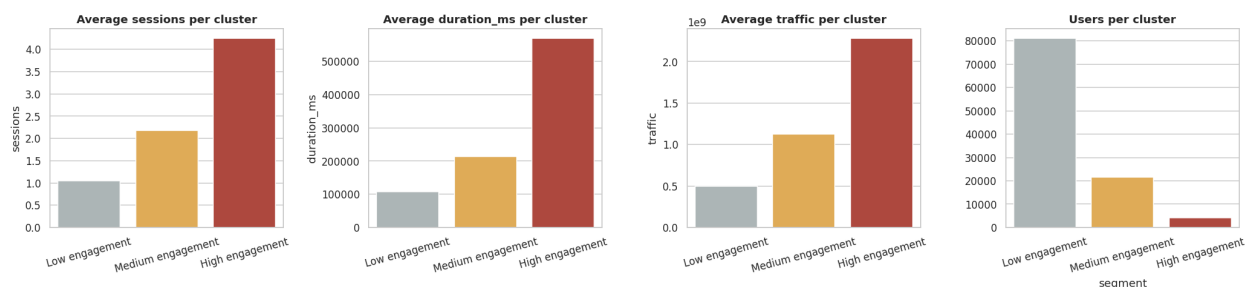


Figure 8 - Average engagement metrics and user counts per cluster

- The elbow method (distance to the line between first and last point) gives **k = 3**, which confirms the choice of three groups.
- Gaming (about 58.3 TB) and Other (about 58.2 TB) make up 96.6% of all application traffic. The top 3 named applications are **Gaming, YouTube and Netflix**. "Other" was left out of the top-3 because it is a catch-all group, not one application.
- Interpretation:** three out of four customers use the network lightly (about 1 session a month). The 4% highly-engaged customers give 12.5% of traffic, so moving customers from low to medium is the biggest volume opportunity.

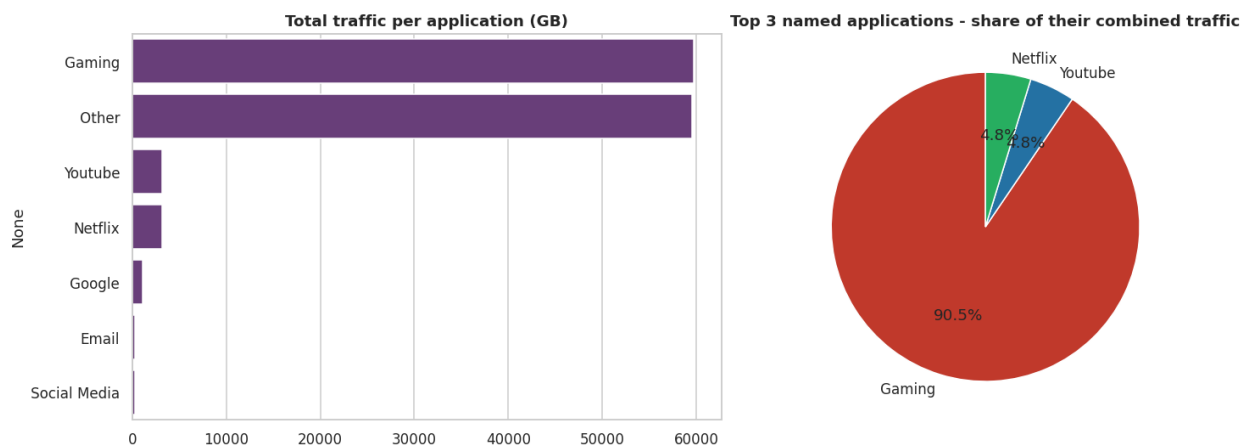


Figure 9 - Total traffic per application and top-3 named applications

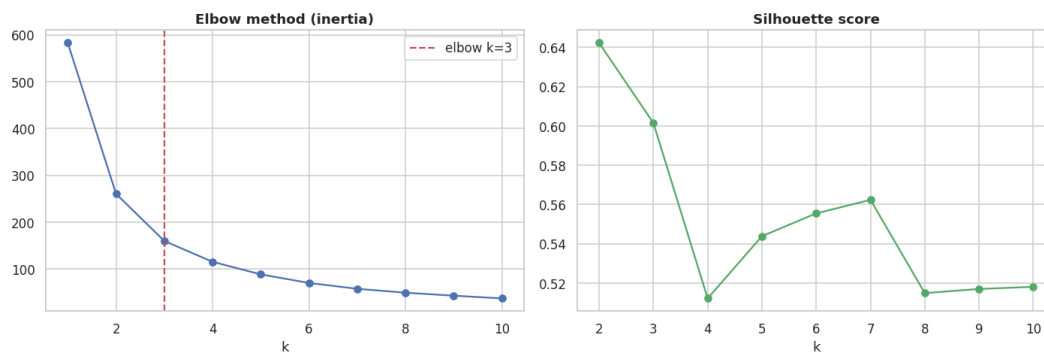


Figure 10 - Elbow method and silhouette score for k-means

Task 3 - User experience

What we did: per customer we calculated average TCP retransmission, average RTT and average throughput (DL + UL) and the most frequent handset type. Missing values were replaced by the mean (numbers) or mode (handset) and outliers by the mean. We listed the top 10, bottom 10 and most frequent 10 values for TCP, RTT and throughput (printed in the notebook), compared handsets, and ran k-means (k=3) on standardised experience metrics.

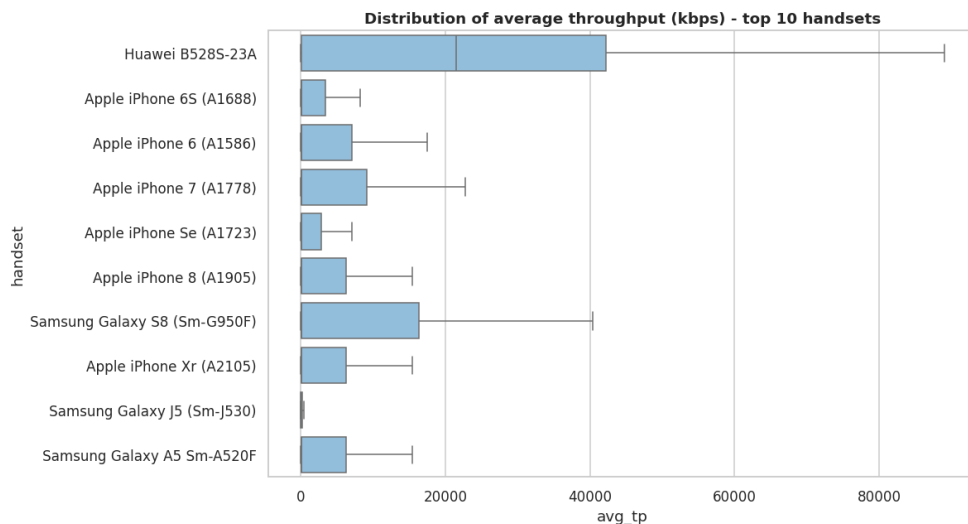


Figure 11 - Distribution of average throughput for the 10 most common handsets

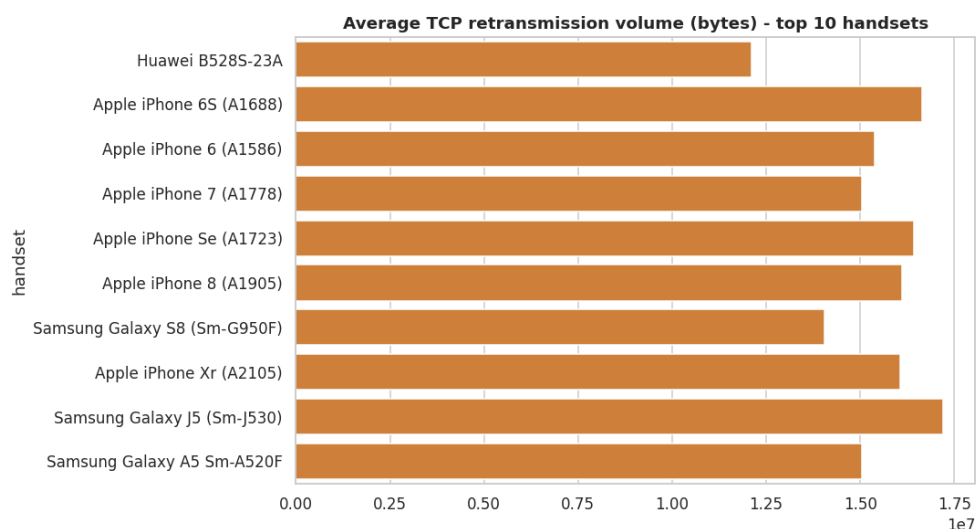


Figure 12 - Average TCP retransmission volume for the 10 most common handsets

- **Throughput:** Huawei routers (E5573, E5180, B528S) average 25-32 Mbps. iPhone 4/4S, ZTE MF259 and Samsung Galaxy J5 are much slower (median for J5 only 78 kbps).
- **Retransmission:** highest for iPhone 6S/7, Galaxy J5 and Samsung G390F (about 17 million bytes per user).

Experience cluster	Users	Avg TCP	Avg RTT	Avg throughput	Meaning
Poor	6,886 (6.4%)	6.3 M	228 ms	16.7 Mbps	Very high delay: laggy browsing and gaming; highest churn risk.
Average	64,824 (60.7%)	20.1 M	72 ms	1.8 Mbps	Normal delay but slow speed and many retransmissions: biggest upgrade chance.
Good	35,146 (32.9%)	4.0 M	74 ms	26.5 Mbps	Fast and clean connection, mostly routers and flagship phones.

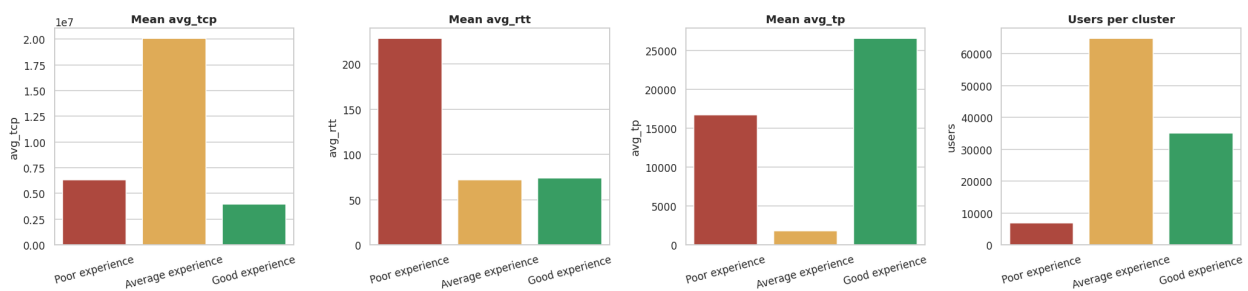


Figure 13 - Experience clusters

Task 4 - User satisfaction

What we did:

- **Engagement score** = Euclidean distance between a customer and the centre of the *least engaged* cluster (from the Task 2 clustering). Further away = more engaged.
- **Experience score** = Euclidean distance between a customer and the centre of the *worst experience* cluster (Task 3). Further away = better experience.
- The two distances are on different scales, so both were rescaled to 0-1 before taking the **satisfaction score = average of both**. (This is our design choice; without it one score would dominate.)
- Regression: 80/20 train-test split with 6 inputs (sessions, duration, traffic, TCP, RTT, throughput). Linear Regression reached $R^2 = 0.76$; Random Forest $R^2 = 0.99$ (RMSE 0.0047, MAE 0.0030).
- k-means with $k=2$ on the two scores, then average satisfaction and experience per cluster.

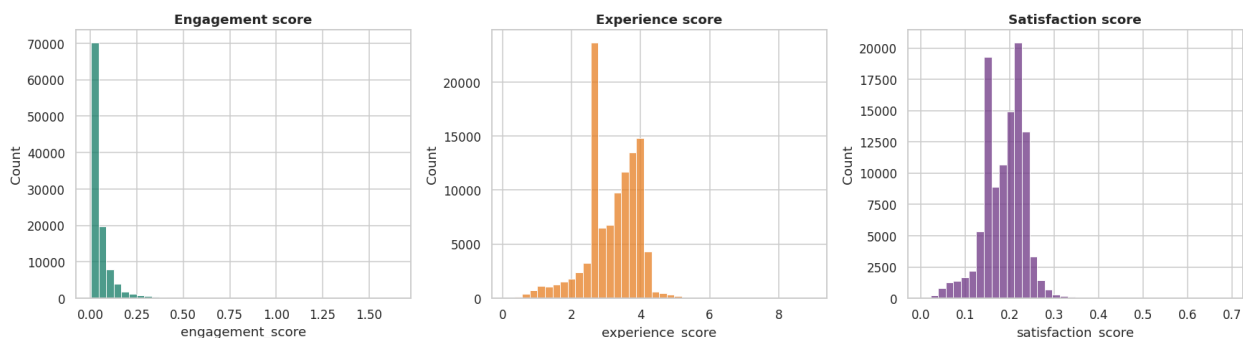


Figure 14 - Distribution of engagement, experience and satisfaction scores

k=2 cluster	Users	Avg satisfaction	Avg experience score
Cluster 0	60,431	0.2185	3.74
Cluster 1	46,425	0.1472	2.43

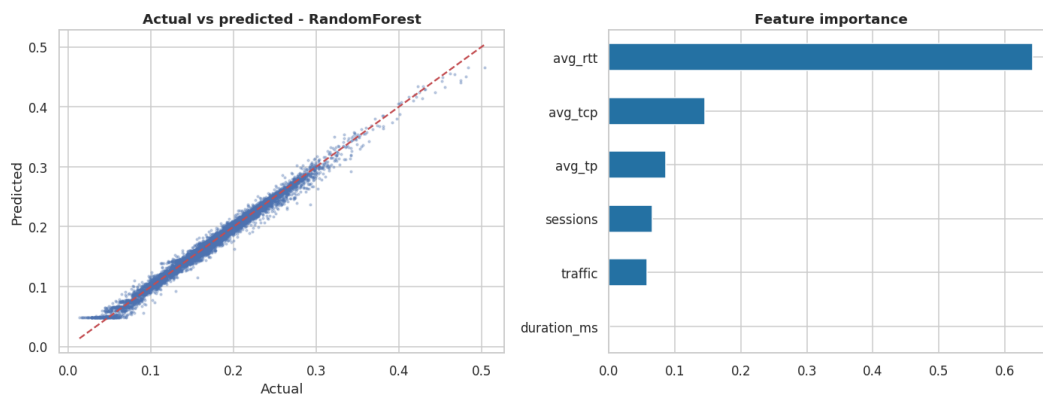


Figure 15 - Regression: actual vs predicted and feature importance

Important note: the R^2 of 0.99 is high because the satisfaction score is calculated from the same inputs. It proves the model reproduces the score, but it does not prove real-life prediction power. A survey-based target would be needed for that.

Export to SQL Server (Task 4.6)

The notebook connects to your local instance **.\SQLEXPRESS** with Windows login, creates the database **telecom_db** if missing, and writes the table **user_satisfaction** (MSISDN, engagement_score, experience_score, satisfaction_score, sat_cluster). It then runs `SELECT TOP 10 * FROM`

user_satisfaction ORDER BY satisfaction_score DESC and saves the output as a picture. In my working environment no SQL Server was available, so the code automatically fell back to a local SQLite file to test the same query; the picture below comes from that test. **On your PC, run the script again and take your own screenshot in SSMS for the final submission.**

SELECT TOP 10 * FROM user_satisfaction ORDER BY satisfaction_score DESC

MSISDN	engagement_score	experience_score	satisfaction_score	sat_cluster
33625779332	1.6417	3.4743	0.6899	0
33614892860	1.4359	2.6971	0.583	1
33626320676	1.3907	2.91	0.5813	1
33659725664	1.22	2.9244	0.5301	1
33675877202	1.2018	2.8956	0.5229	1
33760536639	1.3216	2.058	0.512	1
33650133130	0.014	8.9512	0.5035	0
33650652801	0.0197	8.5792	0.4842	0
33651586339	0.044	8.3946	0.4811	0
33667511187	0.0347	8.34	0.4752	0

Figure 16 - Output of the SELECT query (test run on SQLite fallback)

Model deployment and tracking (Task 4.7)

The final Random Forest is saved as *satisfaction_model.joblib*. Each run appends one row to *model_tracking.csv*: run id, code version (git hash or file hash), start and end time, data source, parameters, metrics (R2, RMSE, MAE) and artifact path. The loss (test MSE) for 5 to 100 trees is saved in *loss_convergence.csv*. If MLflow is installed (`pip install mlflow`), the same information is also logged to MLflow and can be viewed with `mlflow ui`.

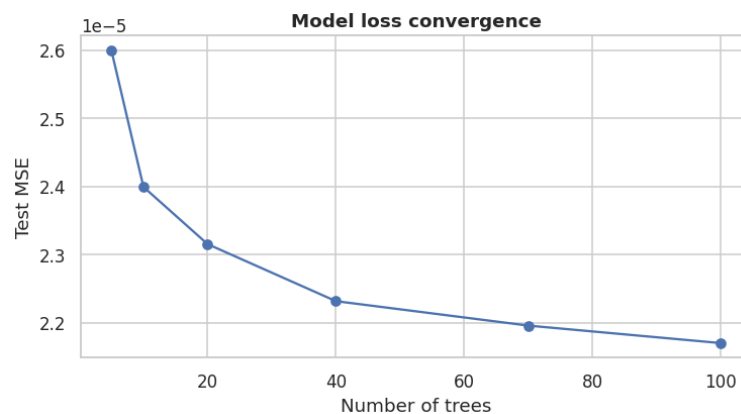


Figure 17 - Loss convergence while adding trees

4. Recommendation

Growth potential: positive. Recommendation: buy TelCo, subject to confirming the financial data.

- **Large upsell base:** 76% of customers are low-engagement. Even moving a part of them to the medium group adds a lot of data traffic.
- **Fixed-wireless / home internet:** a home router is the single most used device and the fastest. Grow this product.
- **Network investment with clear targets:** 6.4% of customers have very high delay and about 60% have low throughput. Fixing them reduces churn and unlocks higher plans.
- **Video and gaming bundles:** Gaming, YouTube and Netflix dominate the traffic. Sell passes and priority speed for them.
- **Handset offers:** Apple has 39.7% of sessions, many on old iPhone 6/6S. Upgrade campaigns are an easy win.

Before the final offer the investor should check revenue, ARPU, churn and costs, because they are not in this dataset.

5. Limitations

- Only one month of data: no trend, seasonality or churn measurement.
- No financial data (revenue, ARPU, cost), so profit is estimated only from usage patterns.
- Application bytes do not add up to the total bytes (about 1.85 times higher); Gaming and Other look unusually large.
- 59-64% of TCP values are missing; mean replacement hides variation and creates a block of identical values.
- Replacing outliers with the mean removes some real heavy users from charts.
- The satisfaction score is a constructed index, not a customer survey; the regression R2 is inflated by design.
- Most customers have only 1-2 sessions in the month, so per-user behaviour is thin.

6. How to run the project on your PC

- Install Python 3.10+ and run: `pip install pandas numpy matplotlib seaborn scipy scikit-learn joblib openpyxl pyodbc sqlalchemy (optional: mlflow)`.
- Install **ODBC Driver 17 for SQL Server** from Microsoft. SQL Server Express must be running (service "SQL Server (SQLEXPRESS)").
- Put *telcom_data__2_.xlsx* in the same folder as *TellCo_Analysis.ipynb*.
- Open *TellCo_Analysis.ipynb* in Jupyter or VS Code and click Run All. The charts are already visible inside the file; a new run also saves them in *charts/* and the tables and model in *outputs/*.
- In SSMS connect to server . \SQLEXPRESS, open database **telecom_db** and run the SELECT query for the screenshot.
- If the connection fails, check the driver name in the SQL_DRIVER line of the script (use "ODBC Driver 18 for SQL Server" if you have version 18).